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Edge AI For HAR

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Edge AI in Action: On-Device Human Activity Recognition

A detailed study on deploying lightweight machine learning models on edge devices.

Abstract

This paper investigates the implementation of Human Activity Recognition (HAR) systems on edge devices. Unlike traditional cloud-based systems, edge AI enables real-time inference directly on devices such as smartphones and embedded systems.

The study evaluates performance trade-offs in terms of latency, accuracy, power consumption, and memory usage.

1. Introduction

Human Activity Recognition (HAR) plays a vital role in applications such as healthcare monitoring, fitness tracking, and smart homes.

Edge AI eliminates dependency on cloud servers by enabling local computation, reducing latency and improving privacy.

Factor	Cloud	Edge
Latency	High	Low
Privacy	Low	High
Bandwidth	High	Low
Energy	High	Optimized

2. Data Acquisition

HAR systems rely on motion sensors such as accelerometers and gyroscopes to collect time-series data.

Sampling frequency and sensor placement significantly impact model performance.

Typical dataset includes activities such as walking, running, sitting, and standing.

System Pipeline

The pipeline shows how data flows from sensors to final predictions.

Edge AI HAR Pipeline

Sensor -> Preprocessing -> Model -> Output

3. Data Preprocessing

Raw sensor data contains noise and inconsistencies that must be cleaned before model training.

Key steps include filtering (low-pass filters), normalization, and segmentation into time windows.

Feature extraction transforms raw signals into meaningful representations such as mean, variance, and frequency components.

Data Processing Flow

Raw sensor data undergoes multiple transformations before being used by ML models.

Data Flow

Raw Data -> Filtering -> Feature Extraction -> Prediction

4. Lightweight Model Design

Due to hardware constraints, models must be computationally efficient.

Algorithms such as Support Vector Machines (SVM), Random Forests, and 1D Convolutional Neural Networks are commonly used.

1D CNNs are particularly effective for time-series data due to their ability to capture temporal patterns.

5. Model Optimization

Model optimization techniques are crucial for deploying AI models on resource-constrained devices.

Technique	Description	Benefit
Quantization	Reduce precision to 8-bit	Smaller size and faster inference
Pruning	Remove redundant weights	Reduced computation
Knowledge Distillation	Transfer learning	Efficient models

Model Optimization

Optimization reduces model size and improves efficiency on edge devices.

Model Optimization

Full Model -> Quantization -> Pruning -> Optimized Model

6. Deployment Pipeline

Models are trained on powerful machines and then converted into edge-compatible formats such as TensorFlow Lite.

Deployment includes integrating the model with sensor data pipelines and running inference locally.

Frameworks like TinyML and Edge Impulse simplify deployment.

7. Evaluation Metrics

Model performance is evaluated using accuracy, precision, recall, and F1-score.

Edge-specific metrics include inference time, memory usage, and power consumption.

Metric	Importance
Accuracy	Correct predictions
Latency	Real-time response
Memory	Hardware constraint
Power	Battery life

8. Conclusion

Edge AI for HAR provides significant advantages in real-time performance and privacy.

Future research directions include federated learning and adaptive edge models.

References

[1] UCI HAR Dataset

[2] TensorFlow Lite Documentation

[3] Edge Impulse

[4] IEEE Research Papers on Edge AI